Handling Data 3 — Measures of dispersion · Quartiles · Cumulative frequency

Quartiles for small data sets: Q_1 is the $\frac{1}{4}(n+1)$th value, the median the $\frac{1}{2}(n+1)$th, Q_3 the $\frac{3}{4}(n+1)$th — if that falls between two values, take their mean. Squared paper for the graph question. Marks in brackets. ★ = harder question.

1 The average number of rainy days per month in a city is: Jan 2, Feb 1, Mar 3, Apr 4, May 15, Jun 24, Jul 27, Aug 26, Sep 20, Oct 12, Nov 5, Dec 2. Calculate **a** the range, **b** the median, **c** the inter-quartile range. [6]

2 Mira divides her garden into nine equal squares and counts the daisies in each: 10, 5, 15, 5, 10, 0, 10, 5, 15. Calculate **a** the range, **b** the median, **c** the inter-quartile range. [6]

3 The marks of 30 boys and 30 girls in a test are grouped below. [13]

Marks x (%)	Boys	Girls
$0 < x \le 20$	3	1
$20 < x \le 40$	7	3
$40 < x \le 60$	10	8
$60 < x \le 80$	6	11
$80 < x \le 100$	4	7

a Draw a cumulative frequency graph for each set on the same axes, and use the graphs to estimate **(i)** the median and **(ii)** the inter-quartile range of each.

b Compare the performance of the boys and the girls.

Total: 25 marks

Check: order the data first · the IQR measures the spread of the middle 50% and ignores extremes · cumulative frequency is plotted at the *end* of each class · for large n read the quartiles at $\frac{1}{4}n$, $\frac{1}{2}n$, $\frac{3}{4}n$ · compare with an average *and* a spread.